Dominion Samuel

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Education

Simon Fraser University ***Burnaby, BC***

Bachelor of Science in Data Science | ***GPA: 4.15*** **Expected Grad Date: December 2027**

- **Courses**: Computational Data Science, Databases, Data Structures and Algorithms, Machine Learning, Artificial Intelligence
- **Awards & Leadership**: President's Honour Roll, Dean's Honour Roll, 2025 Data Science Award, Science Peer Mentor, ColorStack SFU Lead

Experience

Samsung R&D Canada ***Vancouver, BC***

Data Engineering Co-op **January 2026 – August 2026**

- Facilitated the production migration of Apache Airflow from AWS EKS to MWAA, delivering the cutover **3 weeks** ahead of schedule
- Built a custom Cline AI skill to automate Airflow DAG refactoring (v2.4 $\rightarrow$ v3.2) across **25+** repositories, eliminating manual rewrites
- Architected GitHub Actions CI/CD pipelines to package and deploy Airflow DAGs seamlessly across isolated MWAA environments
- Slashed cluster CPU utilization from **90%** to **40%** by diagnosing Argo pod bottlenecks and purging orphaned Airflow metadata
- Decoupled multi-environment MWAA DAG releases, eliminating cross-environment code drift and manual deployment overhead
- Refactored PostgreSQL ETL pipelines by shifting column transformations upstream, accelerating AWS Redshift queries by **12%**
- Tuned PySpark jobs via broadcast joins and dynamic partition pruning, eradicating recurring production dashboard timeouts
- Engineered an MCP server and RAG pipeline over Confluence docs, equipping LLMs with internal context for automated troubleshooting
- Remediated critical SLA bottlenecks in daily batch pipelines, restoring **100%** on-time data delivery to executive dashboards

Projects

High-Throughput Ingestion Pipeline | GitHub | *Python, Polars, Docker, Astro Airflow, GCP* **August 2026**

- Orchestrated a containerized Airflow pipeline with dynamic task mapping to parallelize **130,000+** daily API requests
- Spearheaded rate-limit handling through Airflow connection pools, eliminating **429** Too Many Requests failures at full volume
- Designed a Polars transformation layer to parse nested JSON payloads with strict schema validation into compressed Parquet
- Cut processing time by **40%** by replacing a prior Pandas-based approach with the Polars and Parquet transformation layer
- Pioneered in-memory streaming to Google Cloud Storage via io.BytesIO buffers, removing disk I/O and preventing corrupted uploads

Spotify Music Trends & Hit Prediction | GitHub | *Python, Pandas, NumPy, Scikit-learn, Jupyter Notebooks* **June 2025 – July 2025**

- Extracted acoustic, lyrical, and temporal features from **20+ years** of raw Spotify track metadata using Pandas and NumPy
- Trained K-Means clustering and Random Forest classifiers to predict track popularity, achieving **79%** test accuracy on unseen data
- Optimized model hyperparameters via GridSearchCV and stratified k-fold cross-validation to prevent overfitting and class imbalance
- Quantified key hit predictors including danceability, loudness, and energy through Random Forest feature importance analysis
- Packaged end-to-end data preprocessing pipelines, model weights, and evaluation benchmarks into a reproducible GitHub repository

E-Commerce Analytics Dashboard | GitHub | *SQL, MSSQL, Power BI, Azure* **January 2025 – April 2025**

- Developed MSSQL ETL queries using CTEs, window functions, and aggregates to transform raw transactional data into reporting tables
- Accelerated query performance by **30%** through composite indexing and ETL logic improvements over unoptimized baseline queries
- Formulated custom DAX measures and SQL views to track customer churn, product profitability, and supplier SLA performance
- Created an interactive Power BI dashboard on Azure to surface retention and revenue trends across product, region, and time
- Authored comprehensive **BI documentation** detailing schema design and metric logic to standardize reporting at scale

Technical Skills

Languages: Python, SQL, C++

Data Engineering: ETL/ELT, Data Lakes, Data Warehousing, Apache Spark (PySpark, Spark SQL), Apache Airflow, Polars, Pandas, dbt, Parquet

Cloud & DevOps: AWS (S3, Glue, Lambda, Athena, MWAA, EKS, IAM), GCP (Cloud Storage), Docker, Terraform, Argo, GitHub Actions (CI/CD)

Databases (RDBMS & NoSQL): PostgreSQL, MSSQL, Amazon Redshift, DynamoDB, MongoDB

Reporting & Visualization: Power BI, Tableau, Excel

Developer Tools & AI Agents: Git/GitHub, Cline AI (Prompt Engineering), Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP)